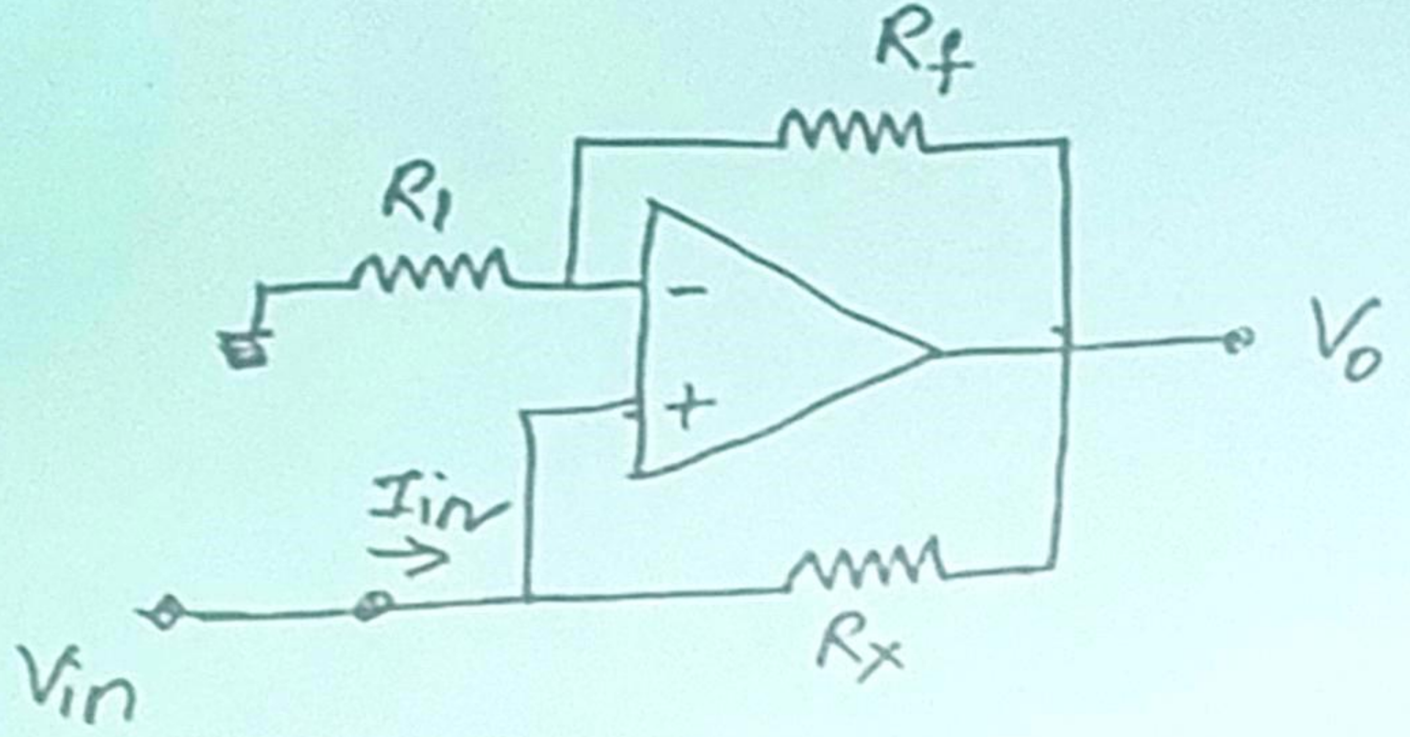
Problem 1

1. Design a two input, two output circuit that yields the sum and difference of inputs, $V_{sum} = V_1 + V_2$ and $V_{diff} = V_1 - V_2$. Try minimizing the component count.

Problem 2

2. Determine $R_{eq} = V_{in}/I_{in}$ of the circuit shown in below Figure.

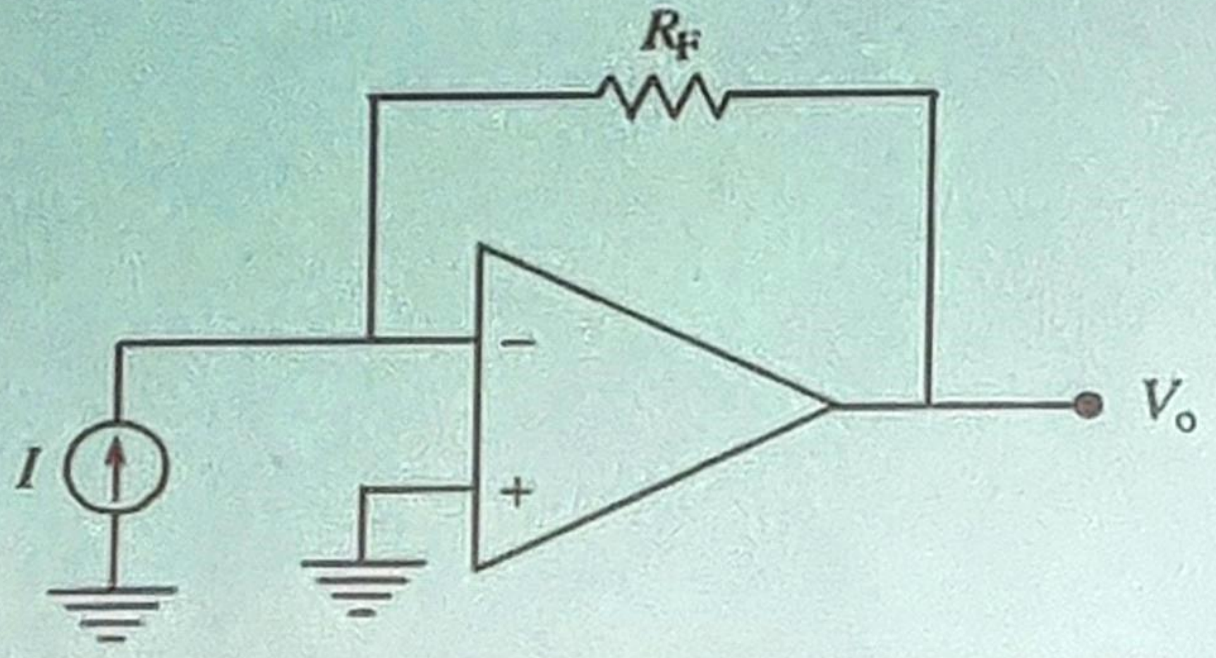
(b) Given $V_{in} = 2V$, $R_x = R_1 = 1k\Omega$, $R_f = 2\ k\Omega$. Obtain R_{eq}.

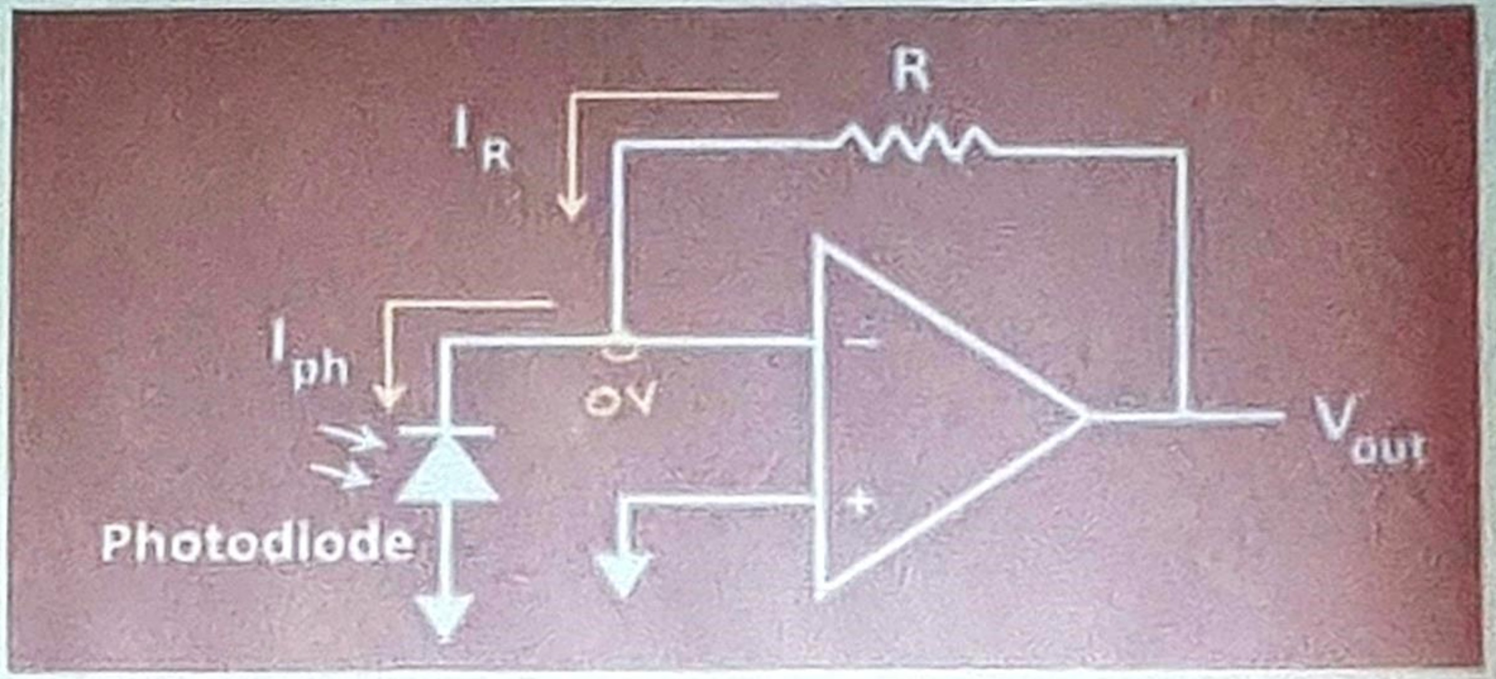


Problem 3

3. Design a circuit to convert 4mA to 20 mA input current to 0V to 8 V output voltage. The reference direction of input current is from ground to circuit

(Transimpedance Amplifier – CCVS – Voltage Shunt Feedback)

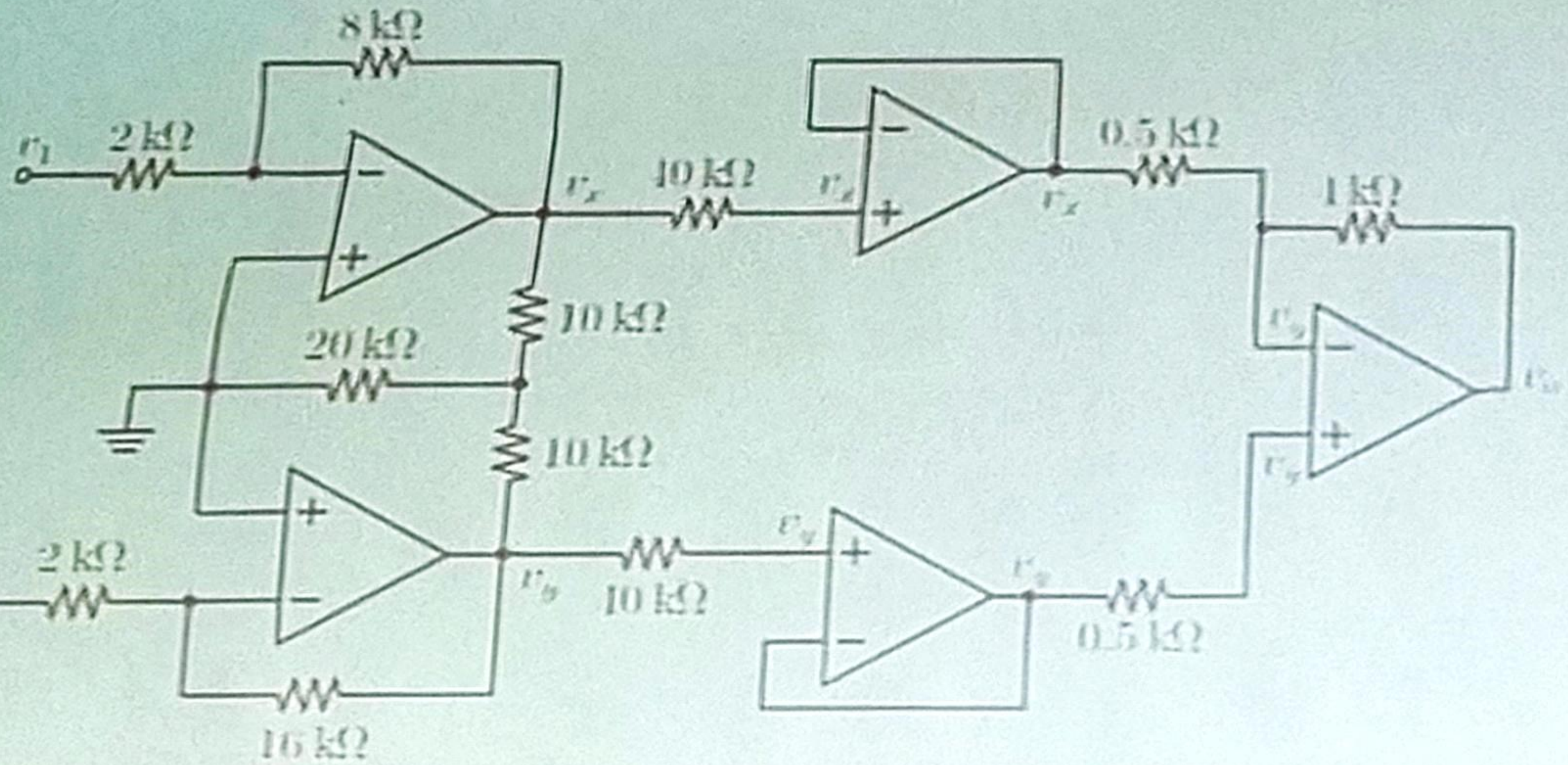




Example : Current to Voltage Converter

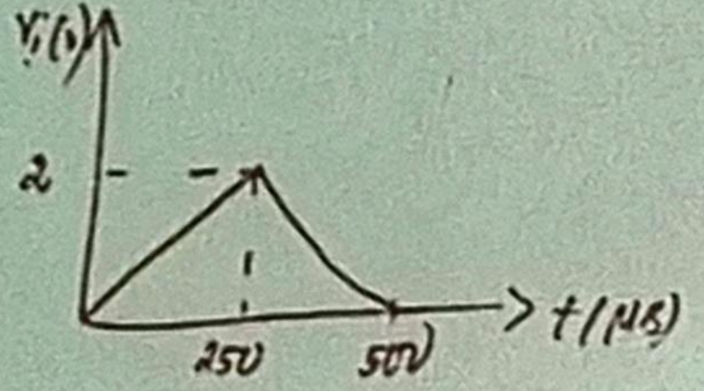
Problem 4

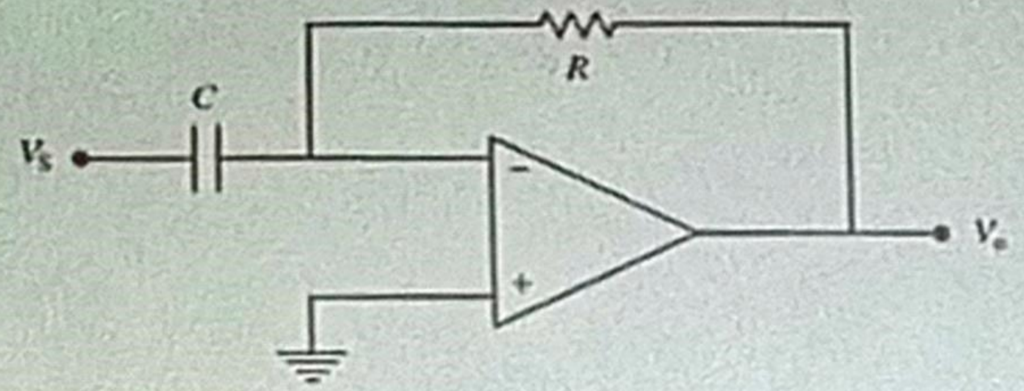
4. If v1 = 1V and v2 = 0.5V, find Vo



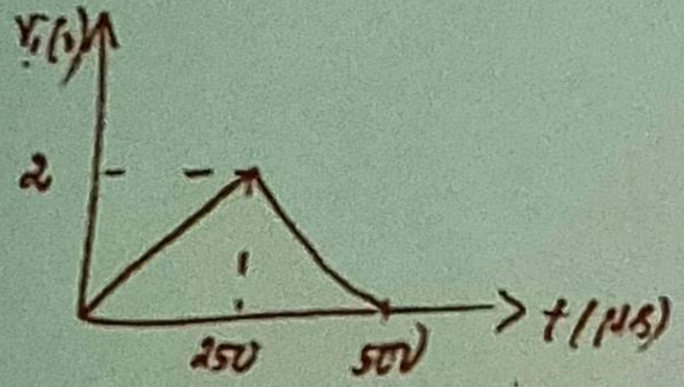
Problem 5

The input to the differentiator circuit is shown in fig. Find the output Vo if $R_f = 2\ k\Omega$ and $C = 0.1\ \mu F$.





5. Solution



for $0 < t < 250\ \mu s$

$V_i(t) = 8000t$ $\left.\begin{array}{l} = 0 \text{ at } t = 0 \\ = 2 \text{ at } t = 250\ \mu s \end{array}\right.$

The output of differentiator $V_0 = -R_f C_1 \frac{d}{dt} V_i(t)$

$V_0 = -2\times10^{3}\times 0.1\times10^{-6} \frac{d}{dt}(8000t)$

$= -1.6V$

Similarly, for 250 μs < t < 500 μs

$V_0 = 1.6V$